

Kernel-checked enclosures of DESI DR2 BAO distances in Lean 4: certifying solver output with integer certificates

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Abstract

We report a small, end-to-end example of certified numerics for cosmology. At $\Omega_m = 0.29743$ (with $hr_d = 101.543$ Mpc), a radiation-off flat Λ CDM fit to the DESI DR2 baryon-acoustic-oscillation data computed with rusty-SUNDIALS (DESI quotes $\Omega_m = 0.2975 \pm 0.0086$ [1]), Lean 4 theorems, checked by the kernel with only the axioms `propext`, `Classical.choice` and `Quot.sound` and without `native_decide`, enclose the dimensionless comoving distance $\chi(z) = \int_0^z dz'/E(z')$ at all seven DESI DR2 effective redshifts with relative width between 2.53×10^{-4} and 2.79×10^{-4} (the Einstein–de Sitter control, a different model, has widths up to 6.6×10^{-4}). The method is elementary: $1/E$ is decreasing, so Riemann sums on a grid bound the integral, and an untrusted program supplies integer bounds of $1/E$ at 4661 grid nodes which the kernel re-checks by exact integer arithmetic. The certified enclosures were then used as a reference to audit a Rust CVMODE reimplementation (rusty-SUNDIALS) and astropy: their values lie inside (a test at the 10^{-4} resolution of the enclosures only; a run at $r_{\text{tol}} = 10^{-3}$ also lies inside); a deliberately loose CVMODE run falls outside at 3 of 7 redshifts; and the enclosures for $\Omega_m \times 1.01$ are disjoint from those of the fit. All hypotheses and controls were preregistered. The paper also records, as research directions, how a survey of the `openai/math` Lean library bears on this programme: it contributes certificate methodology, not results on the underlying physics. Nothing here is a new physical result; the contribution is a verified reference computation and a reusable pipeline.

1 Motivation and scope

Floating-point solvers can fail silently or degrade unnoticed. In the same programme we found that a Rust CVMODE reimplementation had an Adams method stuck at order one (an accuracy loss no test caught) and a Nordsieck-array shift bug that aborted runs at tight tolerances; the commit introducing the faulty rescaling cited, as justification, an unproved Lean axiom whose statement is too weak to certify it. A kernel-checked enclosure removes the solver from the trusted base: if the theorem compiles, the Lean-defined $\chi(z)$ is where the theorem says it is, whatever the solver did; whether that definition is the intended model is a separate, human-checked question.

What is claimed. For $E(\Omega_m, z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$ with $\Omega_m = 29743/100000$ and $z \in \{0.295, 0.510, 0.706, 0.934, 1.321, 1.484, 2.330\}$, the Lean theorems state rational lower and upper bounds for $\chi(z)$ and for $1/E(z)$ (Table 1).

What is not claimed. The fit point is a rounded value of a floating-point minimisation and is not certified as the χ^2 minimum; D_V/r_d , the conversion to D_M/r_d , D_H/r_d and the comparison

with DESI are computed outside Lean from the certified bounds (the pulls in floating point, with diagonal errors only); the model (flat Λ CDM, no radiation) is a modelling choice, not a theorem; no statement about dark energy is made.

2 Method

Model. $\chi(z) = \int_0^z 1/E(\Omega_m, x) dx$ in units of c/H_0 ; $D_M/r_d = \frac{c}{100 hr_d} \chi(z)$ and $D_H/r_d = \frac{c}{100 hr_d} \frac{1}{E(z)}$ with $c = 299792.458$ km/s and $hr_d = 101.543$ Mpc.

Riemann bounds. For $0 \leq \Omega_m \leq 1$, $1/E$ is antitone on $[0, \infty)$ (Lean: `invE_antitone0n`). On the grid $x_i = i/S$ ($S = 2000$), every cell satisfies $f(x_{i+1})/S \leq \int_{x_i}^{x_{i+1}} f \leq f(x_i)/S$ (`cell_bounds`). Every DESI DR2 redshift is a multiple of $1/2000$, so no endpoint is rounded.

Integer certificate. With $\Omega_m = p/q$, $A(x_i) = \Omega_m(1 + x_i)^3 + 1 - \Omega_m = a_i/b$ exactly, with $a_i = p(S + i)^3 + (q - p)S^3$ and $b = qS^3$. For integers ℓ_i, h_i and $D = 10^{12}$, the conditions

$$\ell_i^2 a_i \leq D^2 b \leq h_i^2 a_i$$

imply $\ell_i/D \leq 1/E(x_i) \leq h_i/D$ (`node_bounds`). An untrusted Python program computes $\ell_i = \lfloor D/\sqrt{A(x_i)} \rfloor$ by integer square roots and $h_i = \ell_i + 1$. A Boolean function `checkFrom` tests all 4661 conditions and is evaluated by the kernel (`decide +kernel`); the kernel's arbitrary-precision natural-number arithmetic makes this exact.

Theorem 1 (`BAOCert.seg_bounds`). *Let $p \leq q$, $q, S, D > 0$, and let $L = ((\ell_0, h_0), \dots, (\ell_k, h_k))$ satisfy `checkFrom` $p q S D i_0 L$. Then*

$$\frac{1}{DS} \sum_{j=1}^k \ell_j \leq \int_{i_0/S}^{(i_0+k)/S} \frac{dx}{E(p/q, x)} \leq \frac{1}{DS} \sum_{j=0}^{k-1} h_j.$$

The proof is by induction over the table, using the cell bounds and additivity of the interval integral (Mathlib). For each DESI redshift n/S a generated theorem (e.g. `BAOCert.Fit.chi_z0934`) instantiates it with the prefix of length $n + 1$ and the two sums evaluated by the kernel.

Toolchain. Lean 4.34.1 [2] with Mathlib [3] at commit `d13f23b7`, pinned Mathlib imports, `autoImplicit` off. The core compiles in 9s and the fit-point certificate (table check plus fourteen theorems) in 120s wall-clock (the two control certificates take about the same), one `lean` process on an 8-core virtual machine. Data: `desi_gaussian_bao_ALL_GCcomb_mean.txt` and `_cov.txt` of the DESI DR2 public BAO release [1].

3 Preregistration and controls

Before any table was generated, we committed five hypotheses: H1 (enclosures with relative width $\leq 5 \times 10^{-4}$, standard axioms only), H2 (rusty-SUNDIALS and astropy values inside), H3 (enclosures at $\Omega_m \times 1.01$ disjoint from the fit's for $z \geq 0.5$), H4 (a loose CVODE run, $r_{\text{tol}} = 10^{-2}$, outside at least once) and H5 (a certified χ^2 interval, secondary). Controls: C1, the Einstein–de Sitter closed form $\chi = 2(1 - (1 + z)^{-1/2})$ lies inside the $\Omega_m = 1$ enclosures (checked exactly by squaring); C2, a table with one upper bound lowered by 10^6 units is rejected by the kernel; C3, a `sorry` proof and

a proof from a declared axiom are flagged by the axiom audit. All three controls passed. H1–H4 passed; H5 was not attempted. One deviation: the loose run of H4 also set $a_{\text{tol}} = 10^{-2}$, which the preregistration did not name. An independent verifier recompiled the core and all three certificate files, re-ran the axiom audit, planted its own negative controls (a lowered upper bound is rejected by the kernel; a false stated bound fails to elaborate) and recomputed every enclosure check with 50-digit arithmetic; it confirmed the results with the scope corrections made here.

4 Results

z	n	χ lower	χ upper	rel. width	D_M/r_d	D_H/r_d
0.295	590	0.2747074	0.2747769	2.53×10^{-4}	[8.1104, 8.1124]	[25.42396, 25.42396]
0.510	1020	0.4489509	0.4490704	2.66×10^{-4}	[13.2547, 13.2582]	[22.46849, 22.46849]
0.706	1412	0.5897508	0.5899121	2.74×10^{-4}	[17.4116, 17.4164]	[19.99886, 19.99886]
0.934	1868	0.7341801	0.7343842	2.78×10^{-4}	[21.6757, 21.6817]	[17.47566, 17.47566]
1.321	2642	0.9396161	0.9398783	2.79×10^{-4}	[27.7409, 27.7487]	[14.04071, 14.04071]
1.484	2968	1.0138296	1.0141117	2.78×10^{-4}	[29.9320, 29.9403]	[12.87141, 12.87141]
2.330	4660	1.3157876	1.3161414	2.69×10^{-4}	[38.8469, 38.8574]	[8.63669, 8.63669]

Table 1: Kernel-checked enclosures of $\chi(z)$ and of $1/E(z)$ at the fit point (`formal_cert/BAOCert/Fit.lean`); D_M/r_d and D_H/r_d are these bounds times the exact rational $c/(100 hr_d)$, computed outside Lean.

`#print axioms` on every theorem of the core and of the three certificate files (55 in total) lists only `propext`, `Classical.choice`, `Quot.sound` (the table checks use only `propext`). The certified sources (Core, Fit, FitOm101, EdS) contain no `sorry`, no `axiom` and no `native_decide`; the planted `sorry` and `axiom` of control C3 live in a separate audit file.

z	CVODE 10^{-7}	GK quad.	astropy	CVODE 10^{-3}	CVODE 10^{-2}
0.295	in	in	in	in	in
0.510	in	in	in	in	in
0.706	in	in	in	in	in
0.934	in	in	in	in	out ($+1.37 \times 10^{-4}$)
1.321	in	in	in	in	out ($+4.84 \times 10^{-5}$)
1.484	in	in	in	in	in
2.330	in	in	in	in	out (-2.88×10^{-5})

Table 2: Solver audit against the enclosures (“in”: inside; “out”: outside by the given absolute amount in χ). Columns: rusty-SUNDIALS CVODE at $(r_{\text{tol}}, a_{\text{tol}}) = (10^{-7}, 10^{-10})$ (its default), $(10^{-3}, 10^{-6})$ and $(10^{-2}, 10^{-2})$; its adaptive Gauss–Kronrod quadrature (default); astropy 6.1.7 FlatLambdaCDM($H_0=70$, $\Omega_m=0.29743$, $T_{\text{cmb}}=0$). The rusty-SUNDIALS code is the `qf-bao-distances` crate on the branch of its PR #63.

Table 2 shows the intended use: the certificate grades a solver, and it catches the loose run. Its resolution is the enclosure width: the run at $r_{\text{tol}} = 10^{-3}$ stays inside, so at this width the certificate cannot tell it from the default runs (whose errors the crate’s own tests put at or below about 10^{-6}). H2 is therefore a test at the 10^{-4} level only, and only of χ : the solvers’ $1/E$ values were not compared; second-order bounds (Section 5) would close that gap. The enclosures at $\Omega_m \times 1.01$ are disjoint from the fit’s at every redshift, so the certificate resolves a 1% change in Ω_m .

z	quantity	DESI DR2	σ	prediction from certified bounds	pull
0.295	D_V/r_d	7.9417	–	not certified	–
0.510	D_M/r_d	13.5876	0.1684	[13.2547, 13.2582]	[+1.956, +1.977]
0.510	D_H/r_d	21.8629	0.4289	[22.4685, 22.4685]	[-1.412, -1.412]
0.706	D_M/r_d	17.3507	0.1799	[17.4116, 17.4164]	[-0.365, -0.339]
0.706	D_H/r_d	19.4553	0.3339	[19.9989, 19.9989]	[-1.628, -1.628]
0.934	D_M/r_d	21.5756	0.1618	[21.6757, 21.6817]	[-0.656, -0.619]
0.934	D_H/r_d	17.6415	0.2010	[17.4757, 17.4757]	[+0.825, +0.825]
1.321	D_M/r_d	27.6009	0.3246	[27.7409, 27.7487]	[-0.455, -0.432]
1.321	D_H/r_d	14.1760	0.2246	[14.0407, 14.0407]	[+0.603, +0.603]
1.484	D_M/r_d	30.5119	0.7636	[29.9320, 29.9403]	[+0.749, +0.759]
1.484	D_H/r_d	12.8170	0.5180	[12.8714, 12.8714]	[-0.105, -0.105]
2.330	D_H/r_d	8.6315	0.1011	[8.6367, 8.6367]	[-0.051, -0.051]
2.330	D_M/r_d	38.9890	0.5317	[38.8469, 38.8574]	[+0.248, +0.267]

Table 3: DESI DR2 BAO measurements [1] against predictions from the certified bounds at the fit point (conversion outside Lean); pulls use the diagonal errors only and are computed outside Lean. The D_V/r_d bin is not certified (it needs a cube-root enclosure).

5 Research directions

A read-only survey of the author’s Lean 4 projects against the `openai/math` library (commit `adc7f124`) found no overlap with the K3, moonshine and T-duality core of the programme; the overlap is certificate infrastructure (rational ball arithmetic with a soundness theorem; residual-based matrix-inverse bounds, located by the survey but not line-checked; and integer tables closed by `decide +kernel`, which occurs 1,354,379 times in its `.lean` files while `native_decide` occurs zero times). The present pilot uses that discipline, not that code. The ranked directions are: (R1) a certified DESI DR2 χ^2 interval and a certified exclusion of Einstein–de Sitter, then second-order enclosures (width $\sim 10^{-8}$) and evolving dark energy; (R2) verified integrator theory for rusty-SUNDIALS (BDF order conditions and zero-stability, Nordsieck rescaling), replacing vacuous axioms in that project; (R3) validated Taylor enclosures for a nonlinear ODE with an exact invariant (a 0-D tokamak current quench); (R4) sanity theorems supporting a human audit of Comparator-accepted Lean statements; (R5) removing inconsistent axioms from the existing corpus; (R6) a certified follow-up of a published null result on the matrix Lieb–Thirring inequality [4]. Details are in `docs/RESEARCH_DIRECTIONS_2026-10-08.md`.

6 Limitations

The enclosures are first order: relative width $2.5\text{--}2.8 \times 10^{-4}$ at the fit point and at $\Omega_m \times 1.01$, up to 6.6×10^{-4} for the Einstein–de Sitter control. Only Λ CDM without radiation is covered, at one parameter point. The trusted base is Lean’s kernel and Mathlib; the emitter, the solver runs and the DESI comparison are not trusted and need not be. The rusty-SUNDIALS checkout was a pull-request branch, not its main branch.

Reproducibility and acknowledgements

Lean sources: `formal_cert/BAOCert/`; emitter, evaluation and Rust audit: `scripts/certified_numerics/`; preregistration, compile logs, axiom lists and results: `results/certified_numerics/`

P1_bao/, in the AutoevolveAI repository and in the accompanying bundle. Tables in this paper are generated from those files by `papers/certified_bao/gen_numbers.py`. The work, code and drafting were carried out with the assistance of the AI model Claude (Anthropic) under the author's direction; the author is responsible for the content.

References

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